

Simulation of a Particle Swarm Optimisation MPPT Algorithm

A concise guide to simulation preparation, the live demonstration, and the individual report

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Teaching example: Yingli YLM-J 3.0 PRO 550 W module at standard test conditions

1 The whole idea

An MPPT algorithm makes decisions from measured power. To test those decisions in software, we first need a plant: a mathematical model that tells us what current and power would result from a chosen PV voltage. Without that relationship, the algorithm has nothing meaningful to observe.

Figure 1 shows the complete teaching loop. Datasheet values anchor the PV model. The model produces the current–voltage (I–V) and power–voltage (P–V) curves. P&O or PSO then proposes a voltage reference, receives the resulting voltage and current, calculates power, and makes its next decision.

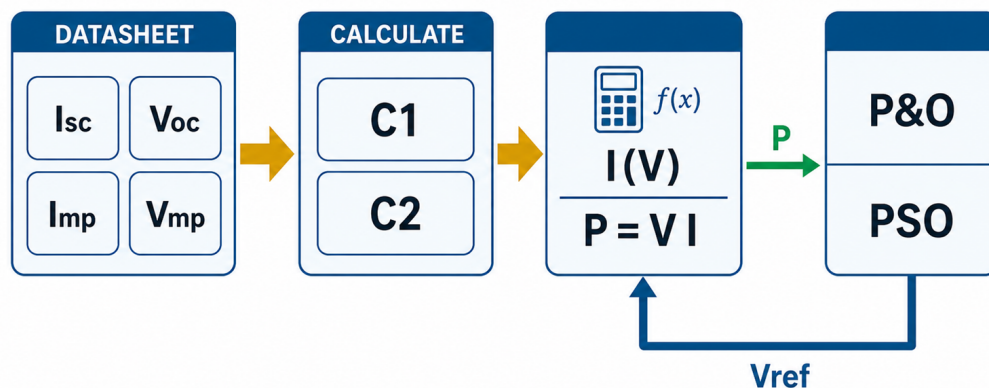


Figure 1: The modelling and MPPT simulation chain. The algorithm receives evaluated power and returns a voltage reference to the empirical plant.

The central idea

The optimiser does not create more solar power. It changes the requested operating point and checks whether the available power increased or decreased.

The simulation repeats four simple actions:

1. apply a candidate PV voltage reference;
2. ask the plant model for current and calculate power;
3. store the result as one settled measurement; and
4. let the MPPT algorithm choose the next candidate, then repeat.

The calculated MPP is shown on plots so that we can judge the algorithm afterward. It is not passed into P&O or PSO. The algorithm receives only its candidate voltage and the corresponding current and power.

Simulation, renderer, and hardware are different. The numerical trace is the simulation result; a table, plot, or GIF only displays it. The scripts assume an ideal inner loop that places the module at the requested voltage. They do not simulate switching, converter transients, a load, sensors, or PWM timing.

2 PV model: C_1 – C_2 parameterisation of a reduced single-diode model

2.1 From the full single-diode equation to an explicit model

A PV module may be represented by the full single-diode equation

$$I = I_{ph} - I_0 \left[\exp\left(\frac{V + IR_s}{A}\right) - 1 \right] - \frac{V + IR_s}{R_{sh}}, \quad A = nN_s \frac{kT_K}{q}. \quad (1)$$

Here, I_{ph} is the light-generated current, I_0 is the diode saturation current, R_s and R_{sh} are the series and shunt resistances, n is the diode ideality factor, N_s is the electrical series-cell count, and T_K is the cell temperature in kelvin. The quantity A has units of volts. It is the module diode-voltage factor and sets the horizontal voltage scale of the exponential. The full equation is implicit because current appears inside its own exponential.

For this practical, two electrical simplifications are applied:

$$R_{sh} \rightarrow \infty, \quad R_s \approx 0. \quad (2)$$

The first neglects shunt-leakage current. The second neglects the internal series-voltage drop and removes current from the exponential. Treating a large R_{sh} as infinite is a recognised analytical simplification in the prescribed single-diode comparison; other formulations retain both parasitic resistances. The resulting explicit reduced model is

$$I(V) = I_{ph} - I_0 \left[\exp\left(\frac{V}{A}\right) - 1 \right]. \quad (3)$$

The simplifications do not claim that the physical resistances are actually zero and infinite. They define a tractable plant for studying MPPT rather than internal module losses. The modules are also assumed uniform, unshaded, at steady state, and at the same irradiance and cell temperature.

At short circuit, $V = 0$, so Equation 3 gives

$$I_{ph} = I_{SC}. \quad (4)$$

This equality is exact within the reduced model; it is not an additional approximation after the series- and shunt-resistance terms have been removed.

2.2 Introduce the normalised C_1 – C_2 form

At the reference condition, define the dimensionless coefficients

$$C_{1,ref} = \frac{I_{0,eff,ref}}{I_{SC,ref}}, \quad C_{2,ref} = \frac{A_{eff,ref}}{V_{OC,ref}}. \quad (5)$$

C_1 normalises the effective diode current by the short-circuit current; C_2 normalises the effective exponential voltage scale by the open-circuit voltage. Rearranging the second definition gives

$$A_{eff,ref} = C_{2,ref} V_{OC,ref}. \quad (6)$$

This does not mean that V_{OC} physically causes the diode-voltage factor. It means that the fitted C_2 expresses the effective exponential voltage scale relative to V_{OC} at the same operating condition.

Substitution into Equation 3 gives

$$I(V) = I_{SC,ref} \left\{ 1 - C_{1,ref} \left[\exp\left(\frac{V}{C_{2,ref} V_{OC,ref}}\right) - 1 \right] \right\}. \quad (7)$$

The practical uses Equation 7 as a *empirical parameterisation of the reduced single-diode shape*. The coefficients reproduce the datasheet knee without separately identifying n , I_0 , R_s , or R_{sh} . Thus, A_{eff} and $I_{0,eff}$ are curve-shape parameters for this teaching model, not independently measured semiconductor parameters. The environmental rules introduced below form a semi-empirical extension of this STC fit.

2.3 Identify the coefficients from datasheet operating points

The selected module is the monofacial Yingli Solar YLM-J 3.0 PRO YL550D-49e 1/2. The manufacturer's [datasheet](#), supplies three operating points at STC: $G_{ref} = 1000 \text{ W/m}^2$, $T_{ref,C} = 25^\circ\text{C}$, and AM 1.5 spectrum.

Table 1: Yingli 550 W operating points used to anchor the STC curve

Operating point	Datasheet values	Role
Short circuit	$V = 0$, $I_{SC} = 13.97 \text{ A}$	Sets the current scale
Open circuit	$V_{OC} = 49.82 \text{ V}$, $I = 0$	Sets the voltage limit
Rated MPP	$V_{MP} = 42.00 \text{ V}$, $I_{MP} = 13.10 \text{ A}$	Locates the knee and gives $P_{MP} = 550.20 \text{ W}$

The endpoints alone do not define the curve between them. The rated MPP supplies the additional shape information. Before using the closed-form expressions, the two conditions that the coefficients are intended to satisfy should be made explicit. At open circuit and at the rated MPP, respectively,

$$1 = C_1 \left[\exp\left(\frac{1}{C_2}\right) - 1 \right], \quad (8)$$

$$1 - \frac{I_{MP}}{I_{SC}} = C_1 \left[\exp\left(\frac{V_{MP}}{C_2 V_{OC}}\right) - 1 \right]. \quad (9)$$

These equations show how C_1 and C_2 control the open-circuit endpoint and the position of the knee.

The prescribed model uses a closed-form approximation. Near the knee and the open-circuit region, the exponential terms are large, so $\exp(x) - 1 \approx \exp(x)$. The resulting coefficient equations are

$$C_2 = \frac{V_{MP}/V_{OC} - 1}{\ln(1 - I_{MP}/I_{SC})}, \quad (10)$$

$$C_1 = \left(1 - \frac{I_{MP}}{I_{SC}}\right) \exp\left(-\frac{V_{MP}}{C_2 V_{OC}}\right).$$

For the selected module, substitution gives $C_{1,ref} = 2.08362 \times 10^{-8}$ and $C_{2,ref} = 0.0565401$. Because the identification uses the large-exponential approximation, the raw equation can leave a negligible residual current at V_{OC} . The implementation therefore enforces the known endpoint $I(V_{OC}) = 0$.

Why use this approximation?

Four familiar datasheet values give a smooth nonlinear plant without an implicit current solver. It is transparent, quick to reproduce, and inexpensive to call many times inside PSO.

2.4 Extend the model to irradiance and cell temperature

The STC model becomes $I(V, G, T_C)$ by updating four connected quantities in sequence: the light-generated current, the effective exponential voltage scale, the open-circuit voltage, and the effective

saturation-current coefficient that closes the open-circuit boundary. Let

$$g = \frac{G}{G_{ref}}, \quad \Delta T = T_C - T_{ref,C}, \quad T_K = T_C + 273.15, \quad T_{ref,K} = T_{ref,C} + 273.15. \quad (11)$$

The datasheet gives the fractional coefficients $\alpha_{Isc} = +0.0005/^{\circ}\text{C}$, $\beta_{Voc} = -0.0027/^{\circ}\text{C}$, and $\gamma_{Pmax} = -0.0035/^{\circ}\text{C}$.

1. Light-generated current. The light-generated current is approximately proportional to irradiance and changes slightly with cell temperature:

$$I_{ph}(G, T_C) = g I_{SC,ref} (1 + \alpha_{Isc} \Delta T). \quad (12)$$

This equation sets the vertical current scale of the I–V curve.

2. Effective exponential voltage scale. The reference scale is recovered from the fitted coefficient:

$$A_{eff,ref} = C_{2,ref} V_{OC,ref} = 2.81683 \text{ V}. \quad (13)$$

Because $A = nN_s k T_K / q$, it is proportional to absolute temperature if n and N_s are assumed constant. The semi-empirical model therefore uses

$$A_{eff}(T_C) = A_{eff,ref} \frac{T_K}{T_{ref,K}}. \quad (14)$$

This equation sets the horizontal voltage scale of the diode exponential.

3. Open-circuit voltage. Open-circuit voltage decreases with increasing cell temperature and changes approximately logarithmically with irradiance. The adopted relation is

$$V_{OC}(G, T_C) = V_{OC,ref} (1 + \beta_{Voc} \Delta T) + A_{eff}(T_C) \ln g. \quad (15)$$

The first term applies the manufacturer's voltage temperature coefficient. The second represents the logarithmic irradiance dependence obtained from the reduced diode equation.

4. Effective saturation-current coefficient. At open circuit, $I = 0$ and $V = V_{OC}(G, T_C)$. Applying this boundary condition to the reduced model gives

$$I_{0,eff}(G, T_C) = \frac{I_{ph}(G, T_C)}{\exp[V_{OC}(G, T_C)/A_{eff}(T_C)] - 1}. \quad (16)$$

Here $I_{0,eff}$ is a closure parameter: it makes the adjusted curve pass through the required open-circuit point. It is not an independently identified material saturation current.

The final irradiance- and temperature-dependent module model is

$$I(V, G, T_C) = I_{ph}(G, T_C) - I_{0,eff}(G, T_C) \left[\exp\left(\frac{V}{A_{eff}(T_C)}\right) - 1 \right] \quad (17)$$

for $G > 0$ and $0 \leq V \leq V_{OC}(G, T_C)$. The corresponding power and maximum-power voltage are

$$P(V, G, T_C) = V I(V, G, T_C), \quad V_{MPP}(G, T_C) = \arg \max_{0 \leq V \leq V_{OC}(G, T_C)} P(V, G, T_C). \quad (18)$$

If the normalised notation is required at a new operating condition, define

$$C_1(G, T_C) = \frac{I_{0,eff}(G, T_C)}{I_{ph}(G, T_C)}, \quad C_2(G, T_C) = \frac{A_{eff}(T_C)}{V_{OC}(G, T_C)}. \quad (19)$$

By definition, $C_2(G, T_C)V_{OC}(G, T_C) = A_{eff}(T_C)$, so this normalised form is algebraically equivalent to Equation 17.

Set current to zero at $G = 0$ and constrain the MPPT simulation to $0 \leq V \leq V_{OC}(G, T_C)$. Clamping current outside the valid voltage interval is a simulation guard, not a physical description of reverse operation. Here T_C is module cell temperature, not ambient temperature. The power coefficient γ_{Pmax} is retained as a separate validation check rather than forced into the current equation.

2.5 Evaluate and verify the model

At STC, the implementation evaluates 1,601 voltage values from zero to 49.82 V. Figure 2 shows the resulting curves.

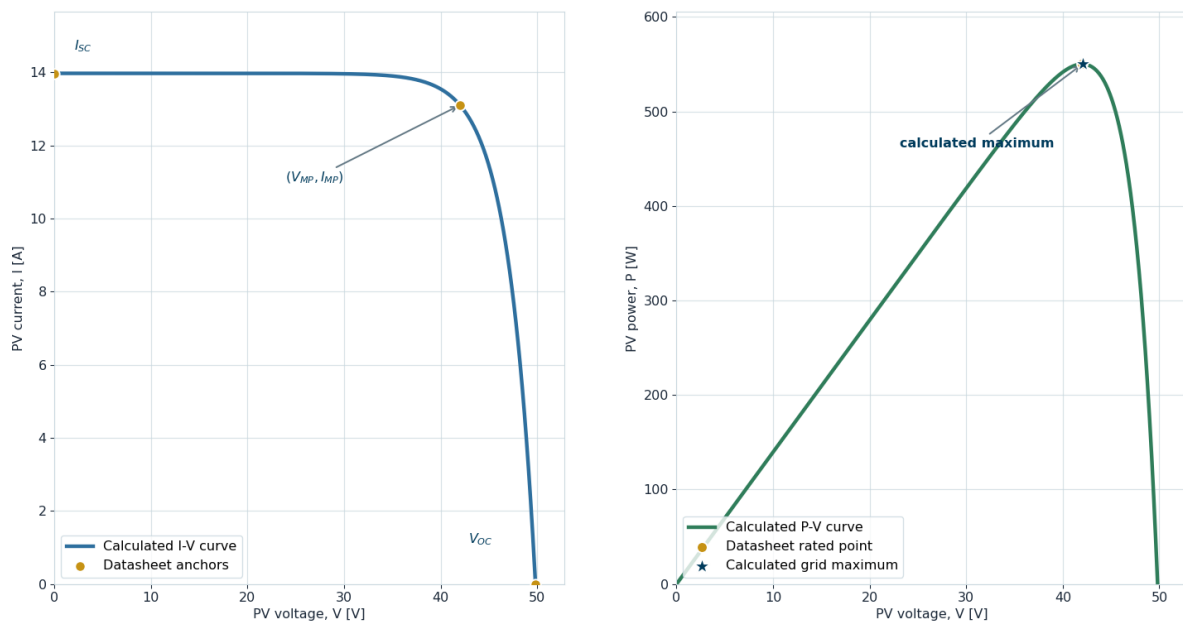


Figure 2: STC I–V and P–V curves from the C1–C2 empirical parameterisation. The rated MPP is an input anchor; the grid maximum checks the implementation.

The model returns $I(42.00 \text{ V}) = 13.1000003 \text{ A}$ and a grid maximum of 42.0356 V, 13.0889 A, and 550.2012 W. Its close STC agreement is expected because the same datasheet anchors identified the coefficients. This is a consistency check, not independent validation of every point between the anchors.

The datasheet’s nominal operating cell temperature (NOCT) table provides a more useful independent check. It reports a second set of electrical values at 800 W/m^2 , 20°C ambient temperature, 1 m/s wind speed, and nominal cell temperature $T_C = 45 \pm 2^\circ\text{C}$. Evaluating the model at $G = 800 \text{ W/m}^2$ and $T_C = 45^\circ\text{C}$ gives:

Table 2: Independent comparison with the Yingli 550 W NOCT data

Quantity	Datasheet	Model	Difference
I_{SC}	11.29 A	11.2878 A	−0.02%
V_{OC}	46.66 V	46.4590 V	−0.43%
I_{MP}	10.48 A	10.4722 A	−0.07%
V_{MP}	39.05 V	38.5610 V	−1.25%
P_{MP}	409.20 W	403.82 W	−1.32%

Agreement within about 1.32% is encouraging for this simplified model, while the NOCT temperature tolerance and the omitted resistive effects must remain part of the interpretation. Students should also plot irradiance and temperature families and check that the directions of change agree with the datasheet coefficients.

Model boundary

The model is explicit and suitable for uniform-condition MPPT simulation. It does not separately represent series loss, shunt leakage, low-irradiance changes in parasitic resistance, mismatch, bypass-diode operation, partial shading, module thermal dynamics, or converter transients. All worked P&O and PSO traces below use STC unless stated otherwise.

Checkpoint before adding an MPPT algorithm

Verify the STC endpoints and rated point; compare the calculated peak with the rated peak; then use the independent 800 W/m² NOCT row to test the environmental extension.

3 Perturb and Observe

P&O is a local search. It nudges the operating voltage, observes the resulting power, and uses that change to choose the next direction. It needs little memory: the previous voltage, previous power, present direction, and step size.

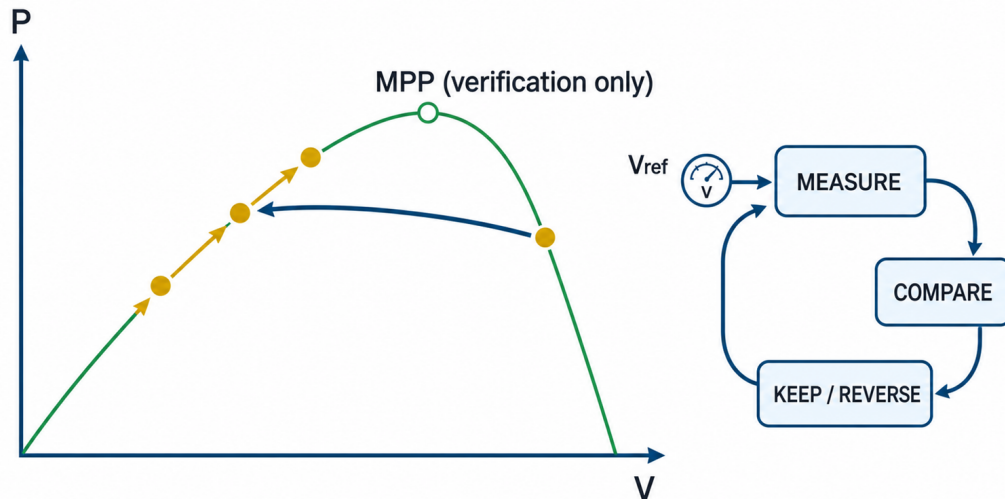


Figure 3: P&O uses consecutive settled measurements. The plotted MPP is a verification reference and is not supplied to the algorithm.

The decision can be written compactly as

$$\begin{aligned} \Delta V &= V[k] - V[k-1], & \Delta P &= P[k] - P[k-1], \\ q_{\text{next}} &= \begin{cases} \text{sign}(\Delta V), & \Delta P \geq 0, \\ -\text{sign}(\Delta V), & \Delta P < 0, \end{cases} \\ V_{\text{ref}}[k+1] &= \text{clip}(V[k] + q_{\text{next}} \Delta V_{\text{step}}). \end{aligned} \quad (20)$$

Table 3: The P&O decision in two rows

Observed change	Interpretation	Next action
$\Delta P \geq 0$	The last move did not reduce power.	Keep the same voltage direction.
$\Delta P < 0$	The last move reduced power.	Reverse the voltage direction.

Listing 1: P&O teaching pseudocode

```

choose Vref, step, and initial direction
measure I and calculate P = Vref * I
repeat:
    apply Vref + direction * step
    measure the new I and P
    if power decreased: reverse direction
    store the new V and P
    clip the next Vref to the allowed range

```

Worked P&O reversal

At 40.5 V the model gives 545.0973 W. A positive probe moves to 43.0 V, where power rises to 547.3560 W. Because $\Delta P = +2.2587$ W, P&O keeps moving upward. The next probe at 45.5 V gives only 498.4960 W, so $\Delta P = -48.8600$ W. P&O now reverses and commands 43.0 V.

With a fixed 2.5 V step, the teaching trace cannot land exactly on 42.0356 V. It moves to either side of the peak and repeatedly corrects itself. This expected hunting illustrates the trade-off: a larger step moves quickly but produces a wider steady oscillation; a smaller step is gentler but takes more measurements to cover the same distance.

Teaching P&O versus the supplied firmware

This Python example directly requests PV voltage and assumes ideal settling. The STM32 implementation perturbs converter gain, waits for the new operating point, uses an eight-sample trimmed mean, rejects small changes with a deadband, adjusts its step size, and finally maps gain to buck and boost duty commands. The example teaches the core decision; it does not reproduce the firmware timing or power stage.

4 Particle Swarm Optimisation

PSO searches with several candidate solutions instead of one local probe. In this one-dimensional example, each particle position is a candidate PV voltage. Its fitness is the power measured at that voltage.

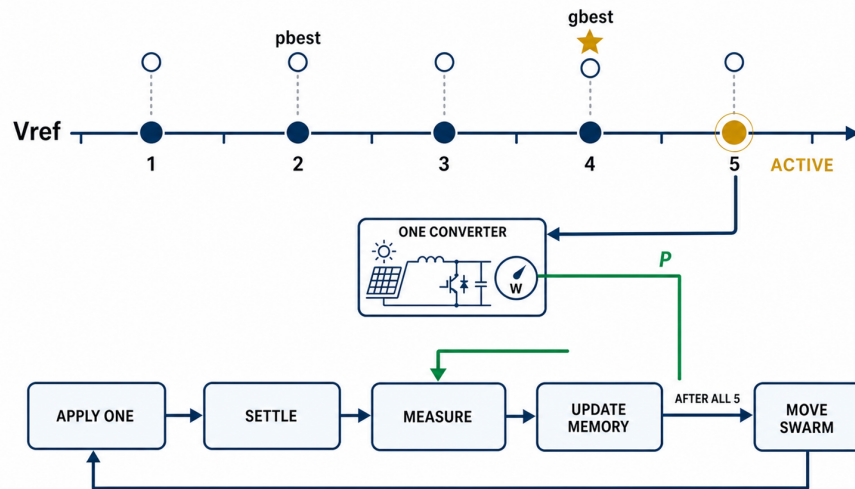


Figure 4: PSO evaluates candidate voltages sequentially, remembers personal and global bests, and then moves the swarm.

Table 4: Minimum PSO vocabulary

Term	Meaning in this simulation
Position x_i	Particle i 's candidate voltage reference
Velocity v_i	Its proposed voltage movement per generation
Fitness	The power $P(x_i)$ obtained at the candidate voltage
Personal best	The best position particle i has measured so far
Global best	The best position measured by any particle so far
Generation	One complete candidate scan followed by one swarm movement

After a generation has been measured, particle i is moved by

$$v'_i = wv_i + c_p r_p (pbest_i - x_i) + c_g r_g (gbest - x_i), \quad x'_i = \text{clip}(x_i + \text{clip}(v'_i)). \quad (21)$$

The three velocity terms have clear roles. Inertia wv_i carries some previous motion forward. The personal term draws the particle toward its own best result. The global term draws it toward the best result shared by the swarm. Random factors r_p and r_g vary the strength of the two attractions.

Listing 2: PSO teaching pseudocode

```

randomly initialise five positions and velocities
repeat for three generations:
  for each particle, one at a time:
    apply Vref, settle, and measure power
    update its pbest and the shared gbest
  if another generation remains:
    update and clip velocities, then move all particles
apply and measure the final gbest once

```

Worked movement for particle 5

After generation 1, particle 5 is at 33.8471 V with velocity +1.8450 V/generation. Its personal best is the same position, while the best measured swarm position is 44.5316 V. Its inertia, personal, and global contributions are +1.1993, 0, and +2.5697 V/generation. The new velocity is therefore +3.7689, which moves the particle to 37.6161 V. Its next 518.5943 W measurement improves its personal best, but not the global best.

The best measured result progresses from 526.9334 W at 44.5316 V in generation 1, to 549.5727 W at 41.5207 V in generation 2, and to 549.9770 W at 42.3108 V in generation 3.

The supplied trace uses seed 40, five particles, three generations, $w = 0.65$, $c_p = c_g = 1.4$, a search range of 4.5–47.0 V, and a 6.0 V/generation velocity limit. It makes 15 sequential candidate measurements, then applies and measures the final global best once: 16 in total.

What the PSO animation represents

The particle positions are updated together in the mathematical generation step, but a single physical converter cannot occupy five operating points at once. Each candidate is applied, allowed to settle, and measured in sequence. The swarm moves only after all five candidates have been tested; the final global best is applied once after generation 3. These deterministic settings were chosen for a clear teaching story. This is conventional PSO, not the PSO–Quasi-Newton method from the prescribed paper.

5 Practical 2: Demonstration

Practical 2 is assessed individually. Attend the slot assigned on ClickUp and be ready to start immediately. Open the correct code, load the required inputs, and have the principal numerical and graphical results available. The slot is for demonstrating and explaining engineering work, not for installing packages, finding files, or repairing an untested environment.

Preparation material and assessed task

The YLM-J 3.0 PRO single-module and P&O examples in this guide prepare you to explain PV modelling and MPPT operation. For the assessed implementation and report, use the same approved YL550D-49e 1/2 module data, construct the assigned string of two identical modules in series, implement the $I(V, G, T_C)$ plant, and solve the maximum-power problem using PSO. A more detailed equivalent-circuit model may support verification, but it does not replace this core task.

The sequence below is indicative; evaluators may change the order or concentrate on any part of the work.

Table 5: Indicative structure of the individual demonstration

Stage	Time	What you must be ready to do
Problem and scope	0–1 min	Identify the PV system, engineering objective, decision variable, constraints, and main assumptions.
Model and code	1–4 min	Locate the module data, single-diode assumptions, C1–C2 identification, environmental update, two-module string calculation, fitness function, and PSO implementation in the code.
Run and results	4–6 min	Run a reproducible case and explain the calculated MPP, curve shape, and PSO convergence rather than merely displaying plots.
Technical questions	6–10 min	Defend the modelling choices, algorithm operation, verification evidence, tool selection, and limitations.

Before your slot

Confirm the ClickUp time and location; test the exact version you will show; keep inputs, code, and results together; record all algorithm settings and random seeds; and know the source, unit, and purpose of every model parameter. Be ready to rerun the simulation or change a setting when asked.

5.1 Questions used to test understanding

The questions are not a fixed script. They test whether you can explain, locate, predict, and justify your work without relying on memorised output values.

Table 6: Representative demonstration questions

Area	Examples
Engineering problem	What is being maximised? What can the optimiser change? What bounds and assumptions apply, and why is MPPT required?
PV model	Which single-diode assumptions lead to the C1–C2 form? How are C_1 and C_2 obtained and updated with G and T_C ? How is the two-module series string formed, and which effects are excluded?
Code and PSO	Where is fitness calculated? What does one particle represent? How do velocity, inertia, personal best, global best, clipping, and the random seed affect the search?
Results	Why is there a peak in the P–V curve? Does the reported voltage, current, and power make physical sense? What change do you predict before a parameter or setting is altered?
Verification	What did you verify, why is the reference valid, how large is the error, and what evidence distinguishes model accuracy from optimiser correctness?
Engineering judgement	Why are the selected tools appropriate? What are their limitations? What changes when sensors, a converter, environmental variation, or module mismatch is included?

6 Practical 2: Report

The report is an engineering argument: define the problem, show how it was modelled and solved, interpret the evidence, and establish why the result should be trusted. It is not a code listing or a chronological account of actions.

Use the following five headings, in this order.

Table 7: Required Practical 2 report structure

Required heading	Minimum content
Energy modelling and optimisation formulation	Open with a short statement of the engineering problem, significance, report purpose, and scope. Give sourced module data with values, units, and relevant tolerances; derive the empirical model, environmental scaling, and series string; then define the objective, decision variable, bounds, constraints, assumptions, and reason for using PSO.
Implementation of the model simulation and solution of the optimisation	Explain the computational workflow and code organisation. State PSO settings, initialisation, bound handling, stopping rule, random seed or run protocol, and software required to reproduce the result. Include only code or pseudocode that supports the explanation.
Simulation results	Present the I–V and P–V behaviour at stated G and T_C , the numerical MPP, environmental sensitivity, PSO convergence, and relevant performance or error measures. Identify the message of each figure or table and explain why it matters.
Summary of engineering methods, skills, and tools applied	Identify each analytical and software tool, its purpose, why it is suitable, and its important limitations.
Integration of engineering methods, skills, and tools for research	Explain how data, theory, implementation, optimisation, verification, and interpretation form one repeatable research workflow. State what would need improvement for a more realistic or hardware-based study.

Do not add a long general introduction before these headings. Begin the first section with one concise paragraph that states the engineering context and ends with a one-sentence roadmap; place equations and detailed assumptions after it.

6.1 Formulate the optimisation explicitly

An optimisation formulation must state the engineering objective, decision variable, objective function, constraints, and assumptions in both mathematics and words. For two identical modules in series at the same irradiance and cell temperature, a voltage-search formulation is

$$\begin{aligned}
 &\underset{V_s}{\text{maximise}} && P_s(V_s, G, T_C) = V_s I_s(V_s, G, T_C), \\
 &\text{where} && I_s(V_s, G, T_C) = I_m \left(\frac{V_s}{2}, G, T_C \right), \\
 &\text{subject to} && 0 \leq V_s \leq 2V_{OC}(G, T_C).
 \end{aligned} \tag{22}$$

Here V_s is the string-voltage decision variable, I_m is the empirical single-module current model at the stated condition, and P_s is the PSO fitness. In words, PSO searches the feasible string-voltage range for the operating point that gives the greatest electrical power available at (G, T_C) . State explicitly that this form assumes identical modules, uniform conditions, no mismatch or shading, and ideal evaluation of each voltage.

What a reproducible formulation records

Give the data source and units, equations and coefficient values, G and T_C , search bounds, PSO coefficients and swarm size, initialisation and seed policy, stopping rule, software environment, and reported outputs. A reader should be able to reproduce the result without guessing an unstated setting.

6.2 How the report is evaluated

Table 8: Practical 2 report mark allocation

Criterion	Evidence expected	Marks
Theoretical and analytical model	Correct sourced data, derivation, units, single-diode assumptions, environmental scaling, series-string model, and optimisation formulation.	15
Software implementation and optimisation	Correct, clear, reproducible model and PSO implementation with justified settings and interpreted outputs.	35
Research-model implementation subtotal		50
Basic verification	Endpoint, unit, rated-value, environmental-direction, range, and physical-plausibility checks.	10
Formal verification	Quantitative comparison with the independent NOCT data, an analytical result, benchmark, published result, or valid alternative method; error measures and a reasoned conclusion about agreement.	10
Diagnostic verification	Convergence, repeated-run, sensitivity, consistency, or related diagnostics with interpretation.	10
Verification subtotal		30
Tool applicability and limitations	Purpose, suitability, assumptions, and limitations of the engineering and software tools.	10
Tool application and integration	A coherent, repeatable workflow linking the methods and tools to the research question and evidence.	10
Methods, skills, and tools subtotal		20
Total		100

Verification must end in a conclusion

For each check, state **what** is verified, **why** the check is needed, **which reference** is used, **how** the comparison is made, and **what the result means**. Verify the PV model and the PSO separately: an optimiser can converge correctly on an inaccurate model. A 20-seed PSO study is one useful diagnostic option, but run count alone is not evidence; interpret the spread, failures, convergence, and sensitivity.

Write for an engineer

Use equations and then explain their physical meaning. Give every axis, value, and parameter a unit. Refer to every figure and table in the text, and state its main message. Report simulated quantities as simulated, not measured. Prefer a short relevant code extract or pseudocode to pages of code. Explain disagreement and limitations rather than hiding them. Finish with conclusions supported by the presented evidence.